

Business Requirements Specifications

WMS Mobile Application

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Business Requirements Specifications

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Business Requirements Specifications

Introduction

The purpose of this document is to describe fully the specification for the solutions associated with the related Business requirements mentioned below.

It is considered as the main reference for the customer, project managers, and functional consultants on how the solutions to the requirements will be reflected in the system or any other component involved.

Using a WMS (Warehouse Management System) mobile app in inventory management streamlines operations by providing real-time data access and updates, reducing errors and improving accuracy. It enables workers to perform tasks like sales, receiving, production, and stock counting efficiently using handheld devices.

1. Summary

This BRS document outlines the functional requirements for a WMS mobile application integrated with SAP B1, focusing on Purchasing, Sales, Production, and Inventory Transactions. It details workflows, data synchronization, and system interactions to ensure seamless operations. The aim is to optimize efficiency and accuracy in warehouse management processes.

2. Business Objectives

Efficient warehouse management is essential to streamline operations, reduce errors, and enhance productivity. The integration between SAP B1 and the WMS mobile application addresses challenges like manual data entry, delays in inventory updates, and lack of real-time visibility into warehouse activities.

3. Effected Modules in SAP Business One

- 1- Items Master Data
- 2- Business Partners Master Data
- 3- Purchasing
- 4- Sales
- 5- Production
- 6- Inventory Transactions
- 7- Batches Management

Business Requirements Specifications

4. Transactions in Scope

- 1- Purchase Orders
- 2- Goods Receipt PO
- 3- Goods Returns
- 4- Sales Delivery Notes
- 5- Sales Returns
- 6- Production
- 7- Issue for Production
- 8- Inventory Transfer Request
- 9- Inventory Transfers
- 10- Counting

5. SAP Business One License Requirements

- 1- One Professional or Indirect Access License.

6. WMS Login Details

Each user will have to access the Application via Email and Password

7. Hardware Requirements

- 1- Android Device Handheld.
- 2- Label Printer.

8. Technical Requirements

- 1- SAP B1 Service Layer
- 2- Dart programming language

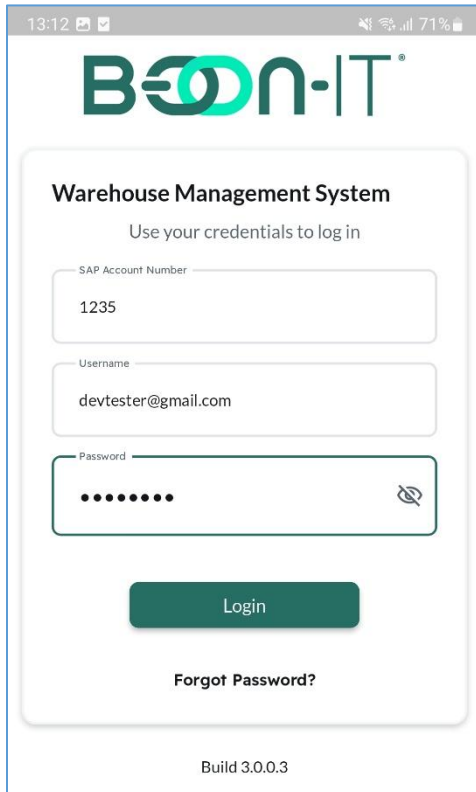
9. Document Control

Date	Author	Position	Version
20-11-2025	Moayed Hribe	Enterprise Solutions Lead	1.0 – Final Document

10. Detailed Business Requirements

10.1 Login Page

The Login page allows authorized warehouse users to securely access the Warehouse Management System (WMS) mobile application using their SAP account number, username, and password.

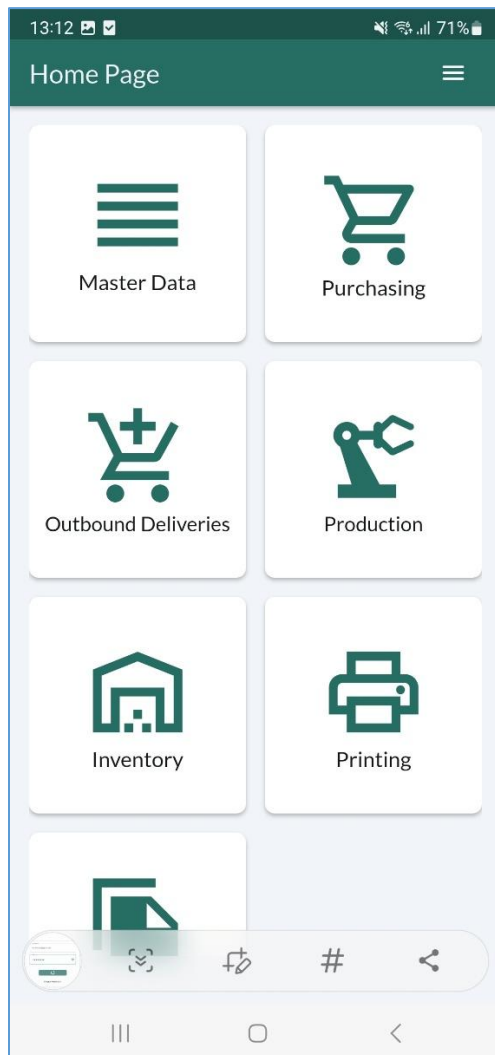
A screenshot of a mobile application login screen. At the top, the status bar shows the time 13:12, signal strength, and 71% battery. The app's logo, 'BEON-IT', is displayed in a stylized green and black font. Below the logo, the title 'Warehouse Management System' is centered, followed by the instruction 'Use your credentials to log in'. There are three input fields: 'SAP Account Number' with the value '1235', 'Username' with the value 'devtester@gmail.com', and 'Password' with masked characters and a toggle icon. A green 'Login' button is positioned below the fields. A link for 'Forgot Password?' is located below the button. At the bottom, the version 'Build 3.0.0.3' is displayed.

- 1- When the user clicks **Login**, the system shall validate:
 - SAP Account Number (Provided By BEON-IT)
 - Username
 - Password
- 2- The system shall authenticate the user against the backend API.
- 3- Successful login shall redirect the user to the **WMS Main Menu**.
- 4- Failed login shall return an error message:
 - Invalid credentials
 - Inactive user
 - Connectivity issues with SAP Business One.

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10.2 Home Page

The Home Page serves as the central navigation hub for users after successful login. It provides quick access to all warehouse operational modules based on the user's permissions.



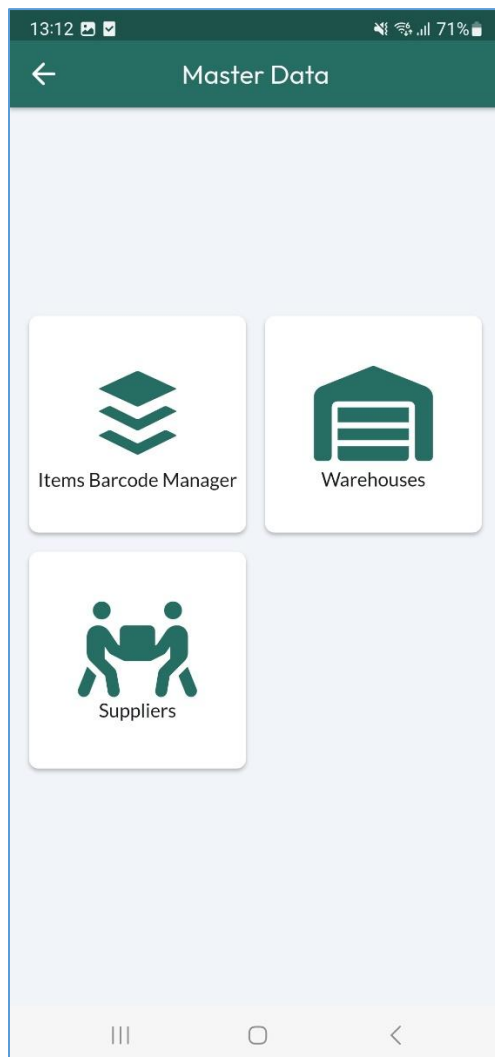
The screen displays:

1. Master Data
2. Purchasing
3. Outbound Deliveries
4. Production
5. Inventory
6. Printing
7. Reports

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10.3 Master Data

The Master Data module provides access to key master records required for all warehouse operations. These records support item barcodes identification, warehouse, and suppliers.



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10.3.1 Barcode Manager

The Barcode Manager page enables users to **manage item barcodes** within the WMS and is fully integrated with SAP Business One. Its main function is to allow the user to:

1. **Select an item** to view its associated barcodes.
2. **Select a Unit of Measure (UoM)** for the barcode being created.
3. **Add a new barcode** for the selected item and UoM.
4. **Edit existing barcodes** assigned to an item.
5. **Delete barcodes** that are no longer needed.
6. **Display all barcodes** related to the selected item (including multiple barcodes).

The screenshot shows a mobile application interface for the Barcode Manager. At the top, there is a status bar with the time 13:12, signal strength, and 71% battery. Below this is a green header bar with a back arrow and the title "Barcode Manager". The main form area has three input fields: "Select Item" with the value "FER0001 - Sulfate Amonium, 25 kg", "Unit" with a dropdown arrow, and "Barcode" with the value "999989". Below these fields is a green "Add" button. At the bottom, there is a table with four columns: "Item Code", "UoM", "Barcode", and "Actions". The table contains two rows of data, each with edit and delete icons in the Actions column.

Item Code	UoM	Barcode	Actions
FER0001	Unit	0000	
FER0001	Unit	1111	

Business Requirements Specifications

10.3.1.2 Barcode Manager – Technical Workflow

1. Item UoM Retrieval

Function

Retrieve all UoMs linked to an item from SAP B1.

Process

1. The system calls a Service Layer SQL Query:
`SQLQueries('GetUOM')/List?ItemCode='<ItemCode>'`
2. SAP returns:
 - UoM Code
 - Base Quantity
 - UoM Entry
3. The application maps this list for display in the UoM dropdown.

2. Creating a New Barcode

Function

Add a new barcode record to SAP Business One using the Service Layer.

Process

1. User selects:
 - Item Code
 - UoM
 - Enters Barcode
2. The system prepares a JSON body using the barcode model.
3. Since this is a new barcode:

`POST /BarCodes`

2. Deleting a New Barcode

Function

Delete an existing barcode in SAP B1.

`DELETE /BarCodes(<AbsEntry>)`

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10.3.1 Warehouses

Function of the Page

The Warehouses page allows the user to:

1. Select a warehouse from the list retrieved from SAP Business One.
2. View warehouse details, including:
 - Warehouse Code
 - Warehouse Name
3. Access two operational reports:
 - Transactions Report (e.g., GRPO, Delivery, Transfers)
 - In-Warehouse Report (stock balance)

This page provides read-only warehouse master data.



The screenshot shows a mobile application interface for the 'Warehouses' page. At the top, there is a dark green header bar with a back arrow on the left and the title 'Warehouses' in the center. Below the header, there is a light blue-grey background area. At the top of this area is a white search bar with the placeholder text 'Select Warehouse'. Below the search bar, the selected warehouse is displayed: '01 - Natagri main warehouse'. Further down, the details are shown: 'Warehouse Code: 01' and 'Warehouse Name: Natagri main warehouse'. At the bottom of the page, there are two dark green buttons with white text: 'Transactions Report' and 'In Warehouse Report'.

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10.3.1.1 Warehouses Technical Workflow

When the user selects a warehouse from the dropdown:

1. The app retrieves the warehouse details.
2. Displays the Warehouse Code and Warehouse Name.

Service Layer Endpoint `GET /Warehouses('<WarehouseCode>')`

Returned Fields

- WarehouseCode
- WarehouseName

`GET /Warehouses('01')`

```
{  
  "WarehouseCode": "01",  
  "WarehouseName": "Natagri main warehouse"  
}
```

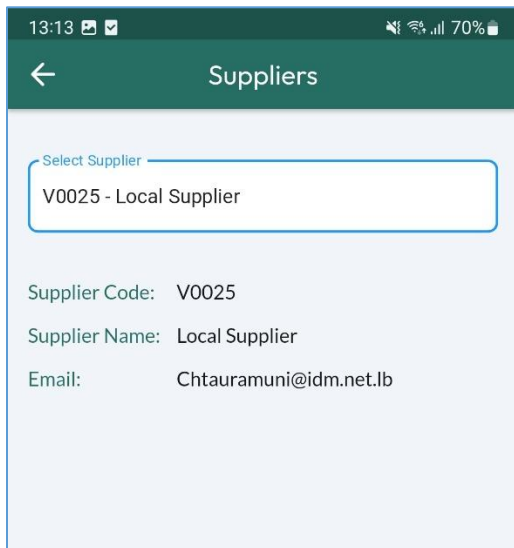
Business Requirements Specifications

10.3.2 Suppliers

The Suppliers page allows the user to:

1. Select a supplier from a list retrieved from SAP B1.
2. Display key supplier information for the selected supplier:
 - Supplier Code
 - Supplier Name
 - Email (and possibly more contact fields later)

The page is read-only, used for quick reference while working in the warehouse.



13:13 70%

Suppliers

Select Supplier

V0025 - Local Supplier

Supplier Code: V0025

Supplier Name: Local Supplier

Email: Chtauramuni@idm.net.lb

10.3.2.1 Suppliers Technical Workflow

Purpose: Show supplier master data when a supplier is chosen from the dropdown.

Service Layer endpoint:

GET /BusinessPartners('V0025')

Business Requirements Specifications

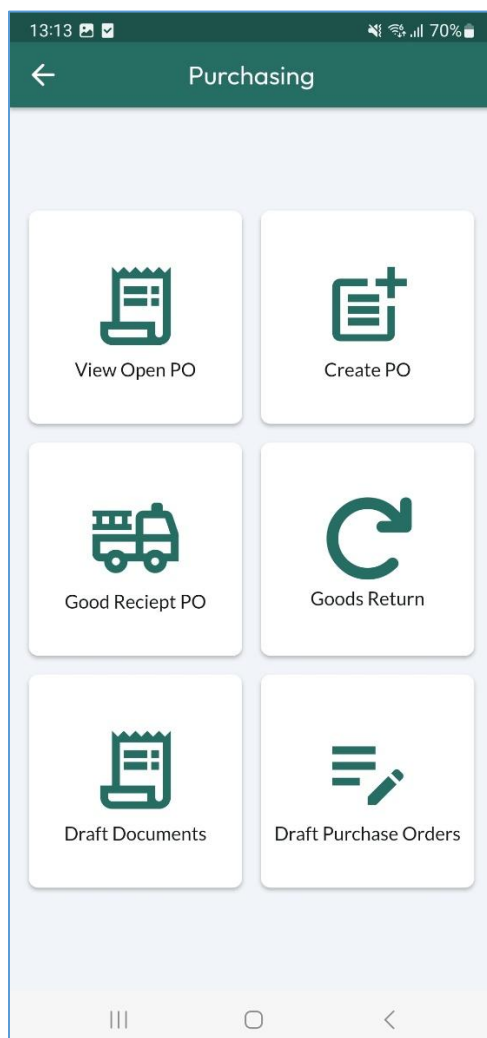
10.4 Purchasing

The Purchasing module allows the warehouse user to perform all procurement-related operations directly from the WMS mobile app.

The page provides access to:

1. View Open Purchase Orders
2. Create Purchase Order (PO)
3. Goods Receipt PO (GRPO)
4. Goods Return
5. Draft Documents (view, edit, post)
6. Draft Purchase Orders (retrieve, update, convert to PO)

These functions allow warehouse and procurement teams to execute purchasing workflows integrated with SAP Business One.



Business Requirements Specifications

10.4.1 Purchase Order

Purpose of the Page

This page allows warehouse or purchasing users to create a new Purchase Order (PO) in SAP Business One directly from the mobile WMS application.

The page supports:

- Selecting a supplier
- Scanning items to add them to the PO
- Entering supplier reference number
- Choosing document dates & warehouse
- Submitting to create the official PO in SAP
- Saving the document as a Draft instead of posting it

The screenshot shows a mobile application interface for creating a purchase order. At the top, the status bar displays the time 13:15, signal strength, and 70% battery. The app header is a dark green bar with a back arrow and the title 'Create Purchase Order'. Below the header, the form contains several input fields: 'V0025 - Local Supplier' (selected), 'Supplier Ref. No.' (empty), 'Scan Item' (0000) with search and camera icons, 'Document Date' (9/11/2025) with a calendar icon, 'Natagri main' (selected warehouse), 'Required Date' (16/11/2025) with a calendar icon, and two buttons: 'Submit' (green) and 'Save as Draft' (grey). At the bottom, a table lists the scanned item.

Barcode	Item Code	Item Name	Q
0000	FER0001	Sulfate Amonium, 25 kg	1

Business Requirements Specifications

10.4.1.1 Purchase Order Technical Workflow

Purpose: Submit PO to SAP B1.

POST /PurchaseOrders

Header JSON Structure

```
{  
  "CardCode": "V0025",  
  "CardName": "Local Supplier",  
  "DocDate": "2025-11-09",  
  "NumAtCard": "Ref No from user",  
  "Comments": "Any comments",  
  "BPL_IDAssignedToInvoice": 1,  
  "DocumentLines": []  
}
```

Lines JSON Structure

```
{  
  "ItemCode": "FER0001",  
  "ItemDescription": "Sulfate Amonium, 25 kg",  
  "Quantity": 1,  
  "WarehouseCode": "01",  
  "UoMEntry": 1,  
  "BarCode": "0000",  
  "UnitPrice": 0,  
  "DiscountPercent": 0  
}
```

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10.4.2 View Open Purchase Order

This page allows the user to view all open Purchase Orders within a selected date range. It helps warehouse staff quickly identify which POs are pending receipt and take action on them.

Date Filter

- The user selects a date (or date range depending on config).
- Clicking Filter retrieves all open POs matching the selected criteria.

Grouping Options

The user can switch between:

- No Grouping
- Group by Supplier
- Group by Date

The list updates instantly to reflect the selected grouping.

Displayed PO Header Information

Each PO entry shows:

- PO Number
- Supplier Name
- Document Date

Expand to View Line Items

When the user expands a PO, the system displays item-level details:

- Item Code
- Item Name
- Ordered Quantity
- Delivery Date

This lets the user confirm what items are expected to be received.

Convert to GRPO

The button "To GRPO" allows the user to:

- Open the GRPO creation screen
- Auto-populated PO lines into the receiving (GRPO) page with ZERO quantity.

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10.4.3 Purchasing

For Receiving the goods, the process starts by warehouse keeper by logging in to the WMS and selecting the supplier start scanning the purchased items, WMS will automatically show the purchase price from the default price list in SAP Business One.

If the item is managed by batch number, the user must define the batch number, batch quantity and expiry date.

In case the Goods Receiving the based-on Purchase order or AP Reserve invoice, the user can choose the base document.

Once the user submits the Purchasing document, WMS will automatically create Goods Receipt PO in SAP Business One.

13:16 70%

Good Reciept PO

V0025 - Local Supplier

PO Number: 202500070 | Date: 2025-11-09T00:00:00

Supplier Ref. No.

Scan Item

Choose Date
9/11/2025

Natagri main

Submit Save as Draft

Item Name	Quantity	Price	UoM	Actions
Sulfate Amonium, 25 kg	0	30.0	Unit	+

Good Reciept PO

VendB - Vendor Branch

☒ Show Purchase Orders ☐ Show Reserve Invoices

Select Purchase Order

Scan Item

Choose Date
8/12/2024

Submit

Head Office

Enter Batch Details

Item Code
MB001

Batch Number
B444-6

Quantity
1

Expiry Date
2024-12-08

Save Cancel

Barcode	Item Code	JoM	Actions
444	MB001	KG	+ -

1-1 of 1

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10.4.3.1 Purchasing Technical Workflow

This section describes how the system creates a Goods Receipt PO (GRPO) using SAP Business One Service Layer, aligned with your **MarketingDocuments**, **DocumentLine**, and **BatchNumber** models.

Header-Level Processing

App Field	SAP Field	Notes
Supplier	CardCode	Required
Supplier Name	CardName	Optional
Document Date	DocDate	Required
Supplier Ref. No.	NumAtCard	Optional
Branch	BPL_IDAssignedToIn voice	If enabled in SAP B1 > Automatic from the assigned branch in warehouse

Lines-Level Processing

App Field	SAP Field	Notes
ItemCode	ItemCode	Required
Item Name	ItemDescription	Optional
Quantity	Quantity	Required
Warehouse	WarehouseCode	Required
UoM Entry	UoMEntry	Needed for correct unit
Barcode	BarCode	Required
Unit Price	UnitPrice	Optional
Batch Numbers	BatchNumbers	If item is batch-managed

Batches - Level Processing

App Field	SAP Field
Batch Number	BatchNumber
Expiry Date	ExpiryDate
Quantity	Quantity
Base Line Number	BaseLineNumber

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JSON Payload Structure Sent to SAP

```
{
  CardCode: "",
  CardName: "",
  DocDate: "",
  NumAtCard: "",
  BPL_IDAssignedToInvoice: "",
  DocumentLines: [
    {
      ItemCode: "",
      ItemDescription: "",
      Quantity: "",
      WarehouseCode: "",
      UoMEntry: "",
      BarCode: "",
      UnitPrice: "",
      BatchNumbers: [
        {
          BatchNumber: "",
          ExpiryDate: "",
          Quantity: "",
          BaselineNumber: ""
        }
      ]
    }
  ]
}
```

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10.4.4 Purchasing Returns

After receiving the purchase and in case there are returns to suppliers, the Warehouse keeper will use WMS application to return the goods back to supplier, the user cannot return goods without predefined Goods Receiving document. In case the returned goods are managed by batches the batches are selected automatically based on the selected Goods Receipt PO

In case the goods receipt PO is closed by AP Invoice in SAP Business One, AP Credit Note must be manually created in SAP Business One by finance department.

Item Name	Quantity	UoM	Actions
Sulfate Amonium, 25 kg	2	Unit	+ -

Business Requirements Specifications

10.4.4.1 Purchasing Returns Technical Workflow

Selecting the GRPO

When the user selects a GRPO, the system loads:

- GRPO Header
- GRPO Lines
- GRPO Batches (if batch-managed)

`GET /PurchaseDeliveryNotes(<DocEntry>)`

Auto-loading the Batches

The application automatically copies the exact same batches numbers used in the GRPO.

For each GRPO line:

You read:

- BatchNumber
- Expiry Date
- Quantity
- BaseLineNumber

Purchase Return Json

```
{  
  "CardCode": "V0025",  
  "CardName": "Local Supplier",  
  "DocDate": "2025-11-09",  
  "Comments": "",  
  "NumAtCard": "",  
  "BPL_IDAssignedToInvoice": 1,  
  "DocumentLines": []  
}
```

Business Requirements Specifications

Purchase Return Lines Json

```
{  
  "ItemCode": "FER0001",  
  "ItemDescription": "Sulfate Amonium, 25 kg",  
  "Quantity": 2,  
  "WarehouseCode": "01",  
  "UnitPrice": 0,  
  "UoMEntry": 1,  
  "BarCode": "0000",  
  "BaseType": "20",  
  "BaseEntry": "GRPO DocEntry",  
  "BaseLine": "GRPO LineNum",  
  "BatchNumbers": [  
    {  
      "BatchNumber": "BATCH-001",  
      "ExpiryDate": "2025-12-31",  
      "Quantity": 2,  
      "BaseLineNumber": 0  
    }  
  ]  
}
```

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10.4.6 Draft Purchasing

The system allows users to save a Goods Receipt PO (GRPO) as a draft instead of posting it to SAP. This enables warehouse users to partially receive items, pause the process, or continue working later without losing progress.

When a user selects Save as Draft, the system stores the GRPO details locally on the device (not in SAP). The draft includes supplier information, document date, scanned items, quantities, and UoM.

Users can later open the Draft GRPO list to review or manage their saved drafts. Each draft displays:

- Draft Number
- Supplier Code
- Document Date

When expanding a draft, the system shows the line details such as:

- Item Code
- Item Name
- Quantity

From this screen, the user can:

1. Copy Draft to GRPO
The selected draft is loaded back into the GRPO creation screen, allowing the user to continue receiving items or finalize the document.
2. Delete Draft
The draft is permanently removed from the device.

Draft #	Supplier Code	Doc Date						
#1	V0025	9/11/2025						
<table border="1"><thead><tr><th>Item Code</th><th>Item Name</th><th>Qty</th></tr></thead><tbody><tr><td>FER0001</td><td>Sulfate Amonium, 25 kg</td><td>13</td></tr></tbody></table>			Item Code	Item Name	Qty	FER0001	Sulfate Amonium, 25 kg	13
Item Code	Item Name	Qty						
FER0001	Sulfate Amonium, 25 kg	13						
<button>Remove Draft</button><button>View GRPO</button>								
#2	V0025	31/8/2025						

Business Requirements Specifications

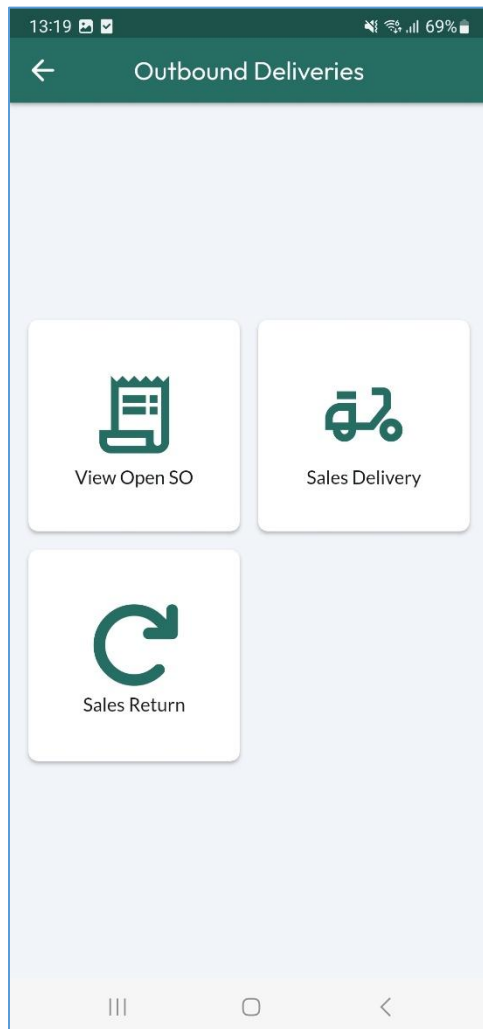
10.5 Outbound Deliveries

The outbound deliveries module allows the warehouse user to perform all sales deliveries-related operations directly from the WMS mobile app.

The page provides access to:

1. View Open Sales Orders
2. Delivery Note
3. Delivery Return

These functions allow warehouse and sales teams to execute sales workflows integrated with SAP Business One.



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10.5.1 View Open Sales Orders

This page allows the user to view all open Sales Orders in SAP Business One based on a selected date filter. The user can filter by document date and optionally group the results by Customer or by Date, or view them without grouping.

The page displays a list of open Sales Orders including:

- Sales Order Number
- Customer Name
- Document Date
- Item details such as Item Code, Item Name, Quantity, and Delivery Date.

Each Sales Order can be expanded to show its item lines.

The user can click “To Delivery” which will open the Sales Delivery transaction with the selected Sales Order information preloaded, allowing the user to continue with the outbound delivery process.

13:20 69%

← All Open SOs

Choose Date

2025-11-09 Filter

No Grouping Group by Customer Group by Date

SO Num	Customer Name	Doc Date
202500234	Liban Agri	2025-11-09

Item Code	Item Name	Quantity	Delivery Date
FER0001	Sulfate Amonium, 25 kg	13.0	2025-11-09T00:00:00Z

To Delivery

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10.5.2 Delivery Notes

The Delivery Note screen allows the user to create a sales delivery directly in SAP Business One.

The user begins by selecting the customer, after which an optional customer reference number can be entered.

Items are then added by scanning their barcodes or selecting them manually, and the application automatically identifies the item and adds it to the list, allowing the user to adjust the quantities as needed.

If the item is batch-managed, the system will automatically retrieve only the available batches for that specific item in the selected warehouse, ensuring that the user can select from accurate stock availability.

The user then selects the posting date and the warehouse from which the items will be delivered.

Once all lines are reviewed and quantities are confirmed, the user clicks Submit, and the application creates a Delivery Note document in SAP Business One through the Service Layer, completing the outbound delivery process.

13:20 69%

← Delivery Note

C0019 - Liban Agri

SO Number : 202500234 | Date -

Customer Ref. No.

Scan Item

Choose Date

9/11/2025

Natagri main

Submit

Item Name	Quantity	Price	UoM	Actions
Sulfate Amonium, 25 kg	13	12.00	Unit	+

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10.5.2.1 Delivery Notes Technical Workflow

This section describes how the system creates a **Delivery Note** using the SAP Business One Service Layer, aligned with the application models: *MarketingDocuments*, *DocumentLine*, and *BatchNumber*.

Header-Level Processing

App Field	SAP Field	Notes
Customer	CardCode	Required
Customer Name	CardName	Optional
Document Date	DocDate	Required
Customer Ref. No.	NumAtCard	Optional
Branch	BPL_IDAssignedToInvoice	Automatically filled from the selected warehouse

Line-Level Processing

App Field	SAP Field	Notes
ItemCode	ItemCode	Required
Item Name	ItemDescription	Optional
Quantity	Quantity	Required
Warehouse	WarehouseCode	Required
UoM Entry	UoMEntry	Required for correct unit conversion
Barcode	BarCode	Required for traceability
Unit Price	UnitPrice	Optional, depends on business logic
Base Type	BaseType	Used only if delivery is based on another document (e.g., Sales Order)
Base Entry	BaseEntry	Required only for copying from base documents
Base Line	BaseLine	Required only for copying from base documents
Row UDFs	UserDefinedFields	Added dynamically
Batch Numbers	BatchNumbers	Only included if the item is batch-managed

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JSON Payload Structure Sent to SAP

```
{
  "CardCode": "",
  "CardName": "",
  "DocDate": "",
  "NumAtCard": "",
  "BPL_IDAssignedToInvoice": "",
  "DocumentLines": [
    {
      "ItemCode": "",
      "ItemDescription": "",
      "Quantity": "",
      "WarehouseCode": "",
      "UoMEntry": "",
      "BarCode": "",
      "BaseType": "",
      "BaseEntry": "",
      "BaseLine": "",
      "BatchNumbers": [
        {
          "BatchNumber": "",
          "ExpiryDate": "",
          "Quantity": "",
          "BaseLineNumber": ""
        }
      ]
    }
  ]
}
```

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10.5.3 Delivery Returns

The Sales Return screen enables the user to return goods previously delivered to a customer.

The user begins by selecting the customer, then choosing the relevant Delivery Note (DN) from which items will be returned. When the Delivery Note is selected, the system automatically retrieves all items and their issued batches as recorded in SAP B1.

The user may scan items to filter quickly or manually adjust the returned quantities. If the item is batch-managed, the application automatically selects and loads the exact batches previously issued in the original Delivery Note to ensure proper reconciliation.

Once the user confirms the return details and submits the transaction, the system creates a Sales Return document in SAP Business One, reversing the quantities and batches from the original issue.

The screenshot shows a mobile application interface for a 'Sales Return' transaction. At the top, the status bar shows the time 13:21 and battery level at 69%. The app header is green with a back arrow and the title 'Sales Return'. Below the header, there are several input fields: a customer selection dropdown showing 'C0019 - Liban Agri', a 'DN Number : 202500234 | Date -' field with a search icon, a 'Scan Item' field with a camera icon, a 'Choose Date' field showing '9/11/2025' with a calendar icon, and a 'Natagri main' dropdown. A green 'Submit' button is located below these fields. The main content area features a table with a green header containing 'Item Name', 'Quantity', 'UoM', and 'Actions'. The table has one data row: 'Sulfate Amonium, 25 kg' with a quantity of '2' and unit 'Unit'. The 'Actions' column for this row contains a green plus icon and a red minus icon. Below the table, there is a pagination indicator '1-1 of 1' and navigation arrows. The bottom of the screen shows standard Android navigation icons.

Item Name	Quantity	UoM	Actions
Sulfate Amonium, 25 kg	2	Unit	 

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10.5.3.1 Delivery Returns Technical Workflow

End Point

POST /b1s/v1>Returns

Header-Level Processing

App Field	SAP Field	Notes
Customer	CardCode	Required
Customer Name	CardName	Optional
Document Date	DocDate	Required
Customer Ref. No.	NumAtCard	Optional
Branch	BPL_IDAssignedToInvoice	If branch management is active
Base Document	BaseEntry (per line)	Auto-filled from selection of DN

Line-Level Processing

App Field	SAP Field	Notes
Item Code	ItemCode	Required
Item Name	ItemDescription	Optional
Quantity	Quantity	Required (return quantity)
UoM Entry	UoMEntry	Matches SAP UoM
Warehouse	WarehouseCode	Required
Barcode	BarCode	Optional
Base Entry	BaseEntry	DN DocEntry
Base Line	BaseLine	DN LineNum
Base Type	BaseType	Always 15 (Delivery Note)

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JSON Payload Structure Sent to SAP

```
{
  "CardCode": "C0019",
  "CardName": "Liban Agri",
  "DocDate": "2025-11-09",
  "NumAtCard": "",
  "BPL_IDAssignedToInvoice": 1,
  "DocumentLines": [
    {
      "ItemCode": "FER0001",
      "ItemDescription": "Sulfate Amonium, 25 kg",
      "Quantity": 2,
      "WarehouseCode": "01",
      "UoMEntry": 1,
      "BarCode": "",
      "BaseEntry": 202500234,
      "BaseLine": 0,
      "BaseType": 15,
      "BatchNumbers": [
        {
          "BatchNumber": "BN-001",
          "ExpiryDate": "2026-05-01",
          "Quantity": 2,
          "BaseLineNumber": 0
        }
      ]
    }
  ]
}
```

Business Requirements Specifications

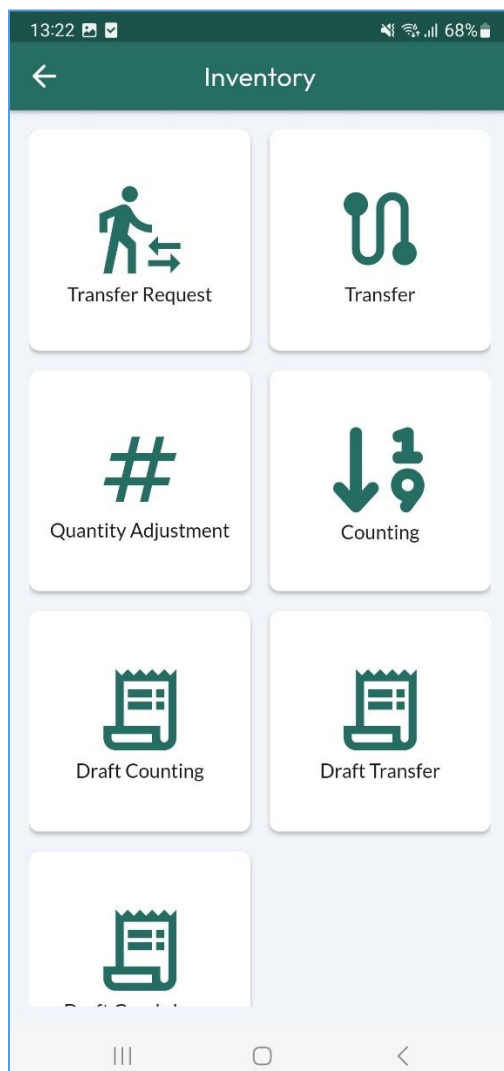
10.6 Inventory

The Inventory module provides users with a complete set of tools to manage internal stock operations directly from the mobile application.

Through this module, warehouse users can initiate Transfer Requests between warehouses, execute real-time stock Transfers, adjust quantities when discrepancies are found, and perform physical Counting activities during stock-taking cycles.

The system also supports saving any counting or transfer process as a draft, allowing users to pause and resume their work without losing progress.

Each function within the module is designed to ensure accurate and efficient stock control, while maintaining alignment with SAP Business One inventory rules such as batch/serial management, warehouse restrictions, and UoM validation.



Business Requirements Specifications

10.6.1 Stock Transfers Requests

The Stock Transfer Request screen allows the user to initiate a request to move inventory from one warehouse to another before performing the actual stock transfer.

The user begins by selecting the source warehouse (From Warehouse) and the destination warehouse (To Warehouse), then chooses the document date.

Items can be added by scanning barcodes or manually searching, the user can adjust quantities and UoM as required.

Once the request is submitted, the application sends the Stock Transfer Request to SAP Business One.

13:22 68%

Transfer Request

From Warehouse
Natagri main

To Warehouse
Fix stock

Choose Date
9/11/2025

Scan Item

Submit

Item Code	Item Name	Quantity	UoM	Action
FER0001	Sulfate Amonium, 25 kg	2	Unit	

Business Requirements Specifications

10.6.1.1 Stock Transfers Requests technical workflow

This section describes how the system creates an Inventory Transfer Request using SAP Business One Service Layer and the models *InventoryTransfer*, *StockTransferLines*.

Service Layer Object

- SAP Object: InventoryTransferRequests
- Endpoint: **POST** /b1s/v1/InventoryTransferRequests

Header-Level Processing

App Field	SAP Field	Notes
From Warehouse	FromWarehouse	Required – source warehouse in SAP
To Warehouse	ToWarehouse	Required – destination warehouse in SAP
Document Date	DocDate	Required
Branch	BPLID	Automatic selection from the selected warehouse

Line-Level Processing

Each row in the grid is mapped to one *StockTransferLines* element in *StockTransferLines*.

App Field	SAP Field	Notes
Item Code	ItemCode	Required
Item Name	ItemDescription	Optional
Quantity	Quantity	Required
From Warehouse	FromWarehouseCode	Required – source warehouse for that line (usually same as header FromWarehouse)
To Warehouse	WarehouseCode	Required – target warehouse for that line (usually same as header ToWarehouse)
UoM Entry	UoMEntry	Required for correct unit

Business Requirements Specifications

JSON Payload Structure Sent to SAP

```
{
  "DocDate": "2025-11-09",
  "DocNum": null,
  "Comments": "Internal transfer request",
  "FromWarehouse": "01",
  "ToWarehouse": "02",
  "BPLID": 1,
  "StockTransferLines": [
    {
      "ItemCode": "FER0001",
      "ItemDescription": "Sulfate Amonium, 25 kg",
      "Quantity": 2,
      "FromWarehouseCode": "01",
      "WarehouseCode": "02",
      "UoMEntry": 1,
      "UseBaseUnits": "N",
      "BatchNumbers": [
        {
          "BatchNumber": "BATCH-001",
          "ExpiryDate": "2026-12-31",
          "Quantity": 2,
          "BaseLineNumber": 0
        }
      ]
    }
  ]
}
```

Business Requirements Specifications

10.6.2 Stock Transfers

The Stock Transfer module allows the user to move inventory from one warehouse to another. The user can optionally select an existing Transfer Request (TR) to load its items automatically, or they may perform a direct stock transfer without any request.

The user selects the *From Warehouse* and *To Warehouse*, chooses the posting date, and then scans or selects items to be transferred.

If the item is batch-managed, the application automatically retrieves only the available batches from the selected “From Warehouse”, ensuring that the user can select valid quantities that physically exist.

Quantities and UoM are editable before submitting. Once submitted, the system posts a Stock Transfer Document into SAP Business One via the Service Layer.

Users can alternatively “Save as Draft,” which saves the transfer locally in the device cache for later review or update before final posting.

13:23

Transfer

TR Number : 10 | Date - 2025-11-09T00:00:00Z

From Warehouse

Natagri main

To Warehouse

Fix stock

Choose Date

9/11/2025

Scan Item

Submit

Save as Draft

Item Name	Quantity	UoM	Actions
Sulfate Amonium, 25 kg	2.0	Unit	+ -

1-1 of 1

Business Requirements Specifications

10.6.2.1 Stock Transfers Technical Workflow

Header-Level Mapping

App Field	SAP B1 Field	Notes
From Warehouse	FromWarehouse	Required
To Warehouse	ToWarehouse	Required
Posting Date	DocDate	Required
Comments	Comments	Optional
Branch	BPLID	If branch management enabled
TR Reference (Optional)	DocNum	Used only when copying from Transfer Request

Lines-Level Mapping

App Field	SAP Field	Notes
Item Code	ItemCode	Required
Item Name	ItemDescription	Optional
Quantity	Quantity	Required
From Warehouse	FromWarehouseCode	Required
To Warehouse	WarehouseCode	Required
UoM Entry	UoMEntry	Required for correct unit
Batch List	BatchNumbers	Only if batch-managed

Batches Mapping

App Field	SAP Field
Batch Number	BatchNumber
Expiry Date	ExpiryDate
Quantity	Quantity
Base Line Number	BaseLineNumber

Business Requirements Specifications

JSON Payload Structure Sent to SAP

```
{
  "DocDate": "2025-11-09",
  "Comments": "Optional",
  "FromWarehouse": "01",
  "ToWarehouse": "02",
  "BPLID": 1,
  "StockTransferLines": [
    {
      "ItemCode": "FER0001",
      "ItemDescription": "Sulfate Amonium, 25 kg",
      "Quantity": 2.0,
      "FromWarehouseCode": "01",
      "WarehouseCode": "02",
      "UoMEntry": 1,
      "UseBaseUnits": "N",
      "BatchNumbers": [
        {
          "BatchNumber": "BT-001",
          "ExpiryDate": "2026-05-10",
          "Quantity": 2.0,
          "BaseLineNumber": 0
        }
      ]
    }
  ]
}
```

Business Requirements Specifications

10.6.3 Quantity Adjustment

The Quantity Adjustment module allows warehouse users to adjust stock levels in SAP Business One based on predefined adjustment reasons such as Expired, Damaged or Waste.

The user selects the adjustment reason and the warehouse, chooses the posting date, and scans or searches for items to adjust. The application retrieves the item details and available batches (if the item is batch-managed) for the selected warehouse.

The user enters or updates the quantity to be reduced and optionally adds remarks. Upon submission, the system creates a Goods Issue transaction in SAP Business One to reduce inventory according to the selected adjustment reason.

Users may also save these adjustments as drafts locally on the device (not pushed to SAP) and later retrieve, update, or delete drafts before final submission.

The screenshot shows a mobile application interface for 'Quantity Adjustment'. At the top, there's a status bar with the time 13:24 and battery level 68%. Below the title bar, there are two dropdown menus: 'Expired' and 'Natagri main'. A date picker shows '9/11/2025'. A 'Scan Item' field with a camera icon is present. Below these are 'Submit' and 'Save as Draft' buttons. A table lists items with columns 'Item Name', 'Quantity', 'UoM', and 'Actions'. The first item is 'Sulfate Amonium, 25 kg' with a quantity of '3' and unit 'Unit'. The table has a pagination bar at the bottom showing '1-1 of 1'.

Item Name	Quantity	UoM	Actions
Sulfate Amonium, 25 kg	3	Unit	

Business Requirements Specifications

10.6.3.1 Quantity Adjustment Technical Workflow

Posted via: Service Layer → *InventoryGenExits*

Header-Level Mapping

App Field	SAP Field	Notes
Adjustment Reason	Comments	Stored in header comments
Warehouse	Warehouse	Required
Posting Date	DocDate	Required
Branch	BPL_IDAssignedToInvoice	Auto-filled based on warehouse branch

Lines-Level Mapping

App Field	SAP Field	Notes
Item Code	ItemCode	Required
Item Name	ItemDescription	Optional
Quantity	Quantity	Required (negative via Goods Issue)
Warehouse	WarehouseCode	Required
UoM	UoMEntry	Required
Batch Numbers	BatchNumbers	Only for batch-managed items

Batches Mapping

App Field	SAP Field
Batch Number	BatchNumber
Expiry Date	ExpiryDate
Quantity	Quantity
Base Line Number	BaseLineNumber

Business Requirements Specifications

JSON Payload Structure Sent to SAP

```
{
  "DocDate": "2025-11-09",
  "Warehouse": "01",
  "Comments": "Expired",
  "BPL_IDAssignedToInvoice": 1,
  "DocumentLines": [
    {
      "ItemCode": "FER0001",
      "ItemDescription": "Sulfate Amonium, 25kg",
      "Quantity": 3,
      "WarehouseCode": "01",
      "UoMEntry": 1,
      "BatchNumbers": [
        {
          "BatchNumber": "BATCH001",
          "ExpiryDate": "2025-12-31",
          "Quantity": 3,
          "BaseLineNumber": 0
        }
      ]
    }
  ]
}
```

Business Requirements Specifications

10.6.4 Stock Counting

The Stock Counting module allows warehouse teams to perform physical inventory counts directly from the mobile application. The user selects the warehouse, chooses the count date, and scans or searches for items, for each item, the user enters the physically counted quantity and, when applicable, selects the batch numbers available in that warehouse.

Depending on the configuration set in the Admin Panel, the system will either:

Creating an Inventory Counting Document

If *Counting* is selected, the mobile app sends the data to SAP B1 as an Inventory Counting document. This records the physical counted quantities but does not affect inventory yet. A manager later reviews and posts the differences inside SAP.

Create an Inventory Posting Document (Direct Stock Adjustment)

If *Posting* is selected, the app sends the data as an Inventory Posting document, which immediately adjusts stock in SAP based on the difference between system quantity and physical count.

Batch-managed items are supported, and only valid batch numbers for the selected warehouse are shown to the user. The document can be saved as a draft or submitted directly, depending on operational workflow.

Item Name	Quantity	UoM	Actions
Sulfate Amonium, 25 kg	150	Unit	+ -

Business Requirements Specifications

10.6.4.1 Stock Counting Technical Workflow

App Setting (Admin Panel)	SAP Document Created	Endpoint Used
Counting	Inventory Counting	/InventoryCountings
Posting	Inventory Posting	/InventoryPostings

Header-Level Mapping

App Field	SAP Field	Notes
Count Date	CountDate (Counting) / PostingDate (Posting)	Required
Branch	BranchID	Optional, depends on SAP setup
Warehouse	WarehouseCode (in each line)	Required
UDFs	UserDefinedFields	Dynamically added

Line-Level Mapping

App Field	SAP Field	Notes
Item Code	ItemCode	Required
Item Name	ItemDescription	Optional
Counted Qty	UoMCountedQuantity	Required
Warehouse	WarehouseCode	Required
UoM Code	UoMCode	Required
Barcode	BarCode	Optional
Batch List	InventoryCountingBatchNumbers / InventoryPostingBatchNumbers	Based on configuration

Batches Mapping

App Field	SAP Field
Batch Number	BatchNumber
Expiry Date	ExpiryDate
Counted Qty	Quantity
Base Line Number	BaseLineNumber

Business Requirements Specifications

JSON Payload Structure Sent to SAP

When Configuration = "Counting"

```
{
  "CountDate": "2025-11-09",
  "BranchID": 1,
  "InventoryCountingLines": [
    {
      "ItemCode": "FER0001",
      "ItemDescription": "Sulfate Amonium, 25 kg",
      "UoMCountedQuantity": 150,
      "WarehouseCode": "MAIN",
      "UoMCode": "Unit",
      "BarCode": "123456789",
      "InventoryCountingBatchNumbers": [
        {
          "BatchNumber": "BATCH01",
          "ExpiryDate": "2026-12-31",
          "Quantity": 150,
          "BaseLineNumber": 0
        }
      ]
    }
  ]
}
```

When Configuration = "Posting"

```
{
  "PostingDate": "2025-11-09",
  "BranchID": 1,
  "InventoryPostingLines": [
    {
      "ItemCode": "FER0001",
      "ItemDescription": "Sulfate Amonium, 25 kg",
      "UoMCountedQuantity": 150,
      "WarehouseCode": "MAIN",
      "UoMCode": "Unit",
      "BarCode": "123456789",
      "InventoryPostingBatchNumbers": [
        {
          "BatchNumber": "BATCH01",
          "ExpiryDate": "2026-12-31",
          "Quantity": 150,
          "BaseLineNumber": 0
        }
      ]
    }
  ]
}
```

Business Requirements Specifications

10.6.5 Production Orders

The Production module enables users to quickly create and execute manufacturing orders based on SAP Business One Production BOMs. Only items that have a Production Bill of Materials in SAP appear in the dropdown list. After the user selects the finished item, warehouse, and production date, the system automatically retrieves the required raw materials from the BOM.

Once the user submits the request, the mobile application performs the complete production cycle automatically, this includes:

1. Creating a Production Order in SAP for the selected finished good.
2. Releasing the Production Order, making it ready for execution.
3. Posting Receipt of Finished Goods into inventory through an *Inventory Goods Receipt* document.
4. Closing the Production Order after completion.

This full automation ensures fast processing, eliminates manual SAP steps, reduces operational errors, and guarantees accurate stock updates for raw materials and finished goods.

The screenshot shows a mobile application interface for 'Production Orders'. At the top, there is a status bar with the time 13:25, signal strength, and 68% battery. Below the status bar is a green header with a back arrow and the title 'Production Orders'. The main form area contains several input fields: 'Select Item' with a dropdown arrow, 'Choose Date' with a calendar icon and the date '9/11/2025', 'Sulfate Amonium, 25 kg' (partially visible), and 'Select Warehouse' with a dropdown arrow. Below these fields are a green 'Submit' button and a green circle with a white plus sign. At the bottom, there is a table with columns: 'Item Code', 'Item Name', 'Quantity', 'Whs', 'Type', and 'Acti'. The table is currently empty. At the very bottom, there is a pagination bar showing '1-0 of 0' and navigation arrows.

Business Requirements Specifications

10.6.5.1 Production Orders Technical Workflow

Production Order Creation (POST /ProductionOrders)

Header-Level Mapping

App Field	SAP Field	Notes
ItemNo	ItemNo	Selected finished item from BOM list
PlannedQuantity	PlannedQuantity	Required production quantity
PostingDate	PostingDate	Date of production
Warehouse	Warehouse	Warehouse for FG receipt

Line-Level Mapping

App Field	SAP Field	Notes
ItemNo	ItemNo	Component item from BOM
PlannedQuantity	PlannedQuantity	Auto-calculated based on BOM × Qty
Warehouse	Warehouse	Component warehouse
Issue Type	ProductionOrderIssueType	e.g., "Manual" / "Backflush"

```
{  
  "ItemNo": "FG-100",  
  "ProductionOrderStatus": "boposPlanned",  
  "PlannedQuantity": 10,  
  "PostingDate": "2025-11-09",  
  "Warehouse": "01",  
  "ProductionOrderLines": [  
    {  
      "ItemNo": "RM-001",  
      "ItemName": "Raw Material One",  
      "PlannedQuantity": 5,  
    }  
  ]  
}
```

Business Requirements Specifications

Production Order Release (PATCH /ProductionOrders(x))

```
{  
  "ProductionOrderStatus": "boposReleased"  
}
```

Receipt of Finished Goods (POST /InventoryGenEntries)

```
{  
  "DocDate": "2025-11-09",  
  "DocType": "dDocument_Items",  
  "DocumentLines": [  
    {  
      "ItemCode": "FG-100",  
      "Quantity": 10,  
      "WarehouseCode": "01"  
    }  
  ]  
}
```

Close Production Order (PATCH /ProductionOrders(x))

```
{  
  "ProductionOrderStatus": "boposClosed"  
}
```

Process Summary

Step	SAP Action	API Endpoint
1	Create Production Order	POST /ProductionOrders
2	Release Production Order	PATCH /ProductionOrders(ID)
3	Post Finished Goods Receipt	POST /InventoryGenEntries
4	Close Production Order	PATCH /ProductionOrders(ID)

Business Requirements Specifications

10.7 Reports

The Reports module gives warehouse users real-time visibility into inventory movement and on-hand stock levels directly from SAP Business One. It consists of two main report types: Transaction Report and In-Warehouse Report, each designed to help users better monitor stock activity and warehouse performance.

The Transaction Report provides a detailed log of all warehouse-related transactions such as Goods Receipt, Goods Issue, Transfers, Counting postings, Production receipts, and Sales-related inventory movements.

The user can filter by item, warehouse, date range, or transaction type.

This enables accurate tracking of inventory flow and helps users quickly identify discrepancies or review historical movements for audit purposes.

The In-Warehouse Report displays the available quantities per item, warehouse. Users can check real-time stock on hand, item availability.

The image displays two side-by-side screenshots of a mobile application interface for SAP Business One reports. Both screens have a dark green header with a back arrow and the report title. The status bar at the top shows the time as 13:28, signal strength, and 67% battery.

Transaction Report (Left Screen):

- Header: Transaction Report
- Filters: Select From Date (8/11/2025), Select To Date (9/11/2025), Natagri main warehouse (dropdown).
- Checkbox: ☐ Filter by item
- Search button
- Table with columns: Document, Base Reference, Date, Item C.
- Table data (7 rows):

Document	Base Reference	Date	Item C
Goods Receipt PO	202500122	2025-11-09	FER00
Goods Issue	202500238	2025-11-09	FER00
Goods Receipt	202500034	2025-11-09	FER00
Goods Issue	202500237	2025-11-09	FER00
Goods Issue	202500237	2025-11-09	FER00
Goods Issue	202500236	2025-11-09	FER00
Inventory Transfer	202500056	2025-11-09	FER00
- Footer: 1-7 of 10, navigation arrows.

In Warehouse Report (Right Screen):

- Header: In Warehouse Report
- Filter: Natagri main warehouse (dropdown)
- Checkbox: ☐ Filter by item
- Checkbox: ☐ Show items with no quantity in stock
- Search button
- Table with columns: Item Code, Item Name, In Stock
- Table data (8 rows):

Item Code	Item Name	In Stock
FER0018	Agromer	12.0
FER0020	Kelpak	1.0
FER0021	Multi N 20 L	5.6
FER0023	Top red 2	2.5
FER0025	Patentkali	32.0
FER0036	Folicist 500 ml	6.5
FER0038	Tioplus 36 K, ...	2.3
- Footer: 15-21 of 182, navigation arrows.

End of Document